

Formulas that may be useful for this exam:

- Limit Laws: If  $\lim_{x \rightarrow a} f(x)$  and  $\lim_{x \rightarrow a} g(x)$  exist,  $c$  is a constant, and  $n$  is a positive integer, then

$$\text{LL-01 } \lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$$

$$\text{LL-02 } \lim_{x \rightarrow a} [f(x) - g(x)] = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x)$$

$$\text{LL-03 } \lim_{x \rightarrow a} [cf(x)] = c \lim_{x \rightarrow a} f(x)$$

$$\text{LL-04 } \lim_{x \rightarrow a} [f(x)g(x)] = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x)$$

$$\text{LL-05 } \lim_{x \rightarrow a} \left[ \frac{f(x)}{g(x)} \right] = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} \text{ if } \lim_{x \rightarrow a} g(x) \neq 0.$$

$$\text{LL-06 } \lim_{x \rightarrow a} [f(x)]^n = \left[ \lim_{x \rightarrow a} f(x) \right]^n$$

$$\text{LL-07 } \lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \rightarrow a} f(x)} \text{ where } n \in \mathbb{N}; \text{ further assume } \lim_{x \rightarrow a} f(x) > 0 \text{ for even } n.$$

$$\text{LL-08 } \lim_{x \rightarrow a} c = c$$

$$\text{LL-09 } \lim_{x \rightarrow a} x = a$$

- Theorem 2.6.5: If  $r > 0$  is a rational number, then  $\lim_{x \rightarrow \infty} \frac{1}{x^r} = 0$ ;

furthermore, if  $x^r$  is defined for all  $x$ , then  $\lim_{x \rightarrow -\infty} \frac{1}{x^r} = 0$ .

- Precise definition of a limit:

$$\lim_{x \rightarrow a} f(x) = L \in \mathbb{R} \Leftrightarrow \forall \epsilon > 0, \exists \delta > 0 \text{ s.t. } 0 < |x - a| < \delta \Rightarrow |f(x) - L| < \epsilon.$$

- Precise definition of a limit for the case  $x \rightarrow -\infty$ :

$$\lim_{x \rightarrow -\infty} f(x) = L \Leftrightarrow \forall \epsilon > 0, \exists \text{ a corresponding number } N \text{ s.t. if } x < N \Rightarrow |f(x) - L| < \epsilon.$$

$$\bullet \quad f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

✓ 1. (Retrieval Practice) Briefly (in one or two lines) describe the key concept of the definition or content for the following items:

1

- 1a. [5pts] function:
- 1b. [5pts] the process of constructing an **inverse function**:
- 1c. [5pts] right-hand limit:
- 1d. [5pts] function  $f(x)$  is continuous at  $x = a$ :
- 1e. [7pts] intermediate value theorem:
- 1f. [7pts] squeeze theorem

2. ~~1a.~~ [6pts] Find the domain and range of  $f(x) = (9 - x^2)^{1/2}$ .

~~1b.~~ [6pts] Find the domain of  $f \circ g$ , where  $f(x) = x^{-1/2}$  and  $g(x) = x - 1$ .

✓ 2c. [6pts] Simplify the expression  $\cos(\tan^{-1}(x))$ .

✓ 2d. [7pts]  $\lim_{x \rightarrow (\pi/2)^+} x^{-1} \sec x = ?$

✓ 2e. [7pts] Find the derivative of  $f(x) = 1/\sqrt{x}$  at the number  $a$ .

✓ 2f. [8pts] If  $f(x) = \ln(e^x - 3)$ , find  $f^{-1}$ , and then compare the domain of  $f$  and  $f^{-1}$ .

3. ~~3a.~~ [8pts] Determine  $\lim_{h \rightarrow 0} \frac{\sqrt{h^2+12}-2\sqrt{3}}{h^2}$ . Any related Limit Law should be indicated in steps.

~~3b.~~ [8pts] Determine  $\lim_{x \rightarrow -\infty} \frac{3x^3+25x^2+121}{\sqrt{3x^3-5x-11}}$ . Any related Limit Law should be indicated in steps.

✓ 3c. [10pts] Determine  $\lim_{x \rightarrow \infty} \frac{\sin x}{x}$  exist or not.

✓ 3d. [10pts] Sketch the graph of  $f(x)$  and its asymptotes satisfying all of eight conditions:  $\lim_{x \rightarrow 0} f(x) = 0$ ,  $\lim_{x \rightarrow \infty} f(x) = 2$ ,  $\lim_{x \rightarrow -\infty} f(x) = 4$ ,  $\lim_{x \rightarrow -4} f(x) = 2$ ,  $\lim_{x \rightarrow -2^-} f(x) = \infty$ ,  $\lim_{x \rightarrow -2^+} f(x) = -\infty$ ,  $\lim_{x \rightarrow 1} f(x) = \infty$ , and  $\lim_{x \rightarrow 2} f(x) = 3$ . [Hint: Apply the definitions of asymptotes and limits. Using only the point-plotting method will result in zero credit.]

3e. [10pts] Use precise definition as  $x$  approaches to  $-\infty$  to find  $\lim_{x \rightarrow -\infty} \frac{1}{x}$ .

4. Let  $f(x) = \begin{cases} 0 & \text{if } x \text{ is rational;} \\ 1 & \text{if } x \text{ is irrational.} \end{cases}$

Prove or disprove:

(You must provide the specific evidence, your reasoning and/or a counterexample. Answering only yes or no will result in zero points.)

4a. [5pts] If  $a$  is a rational number, then there is an irrational number  $b$  satisfying  $|a - b| \leq \frac{1}{10^n}$  for any given  $n \in \mathbb{N}$ .

4b. [5pts] If  $c$  is an irrational number, then there is a rational number  $d$  satisfying  $|c - d| \leq \frac{1}{10^n}$  for any given  $n \in \mathbb{N}$ .

4c. [10pts] The Precise Definition of a limit can be used to prove whether the  $\lim_{x \rightarrow 0} f(x)$  exists.

4d. [5pts]  $f(x)$  is a continuous function. [Hint: Apply 4a and 4b.]

4e. [5pts] Applying the Intermediate Value Theorem to the 4d, we can obtain the same conclusion.

[End of Question Page]

## 國立臺灣海洋大學

## 考試試卷

考題隨試卷繳回

科目 微積分

學系 資工

年級 一

記

分

教師簽名或蓋章

2025年 10月 13日

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(中文)

從此處開始寫起。試卷用紙務須節用。非經主試認可不得續用其他紙張作答。

記分欄

1a. 2 集合  $A, B$ , 每一個  $A$  中的元素, 都能有一個  $B$  的元素對應的關係 <sup>T5</sup>1b. 將原函數對應域的所有元素對應至原定義域,  $f^{-1}(f(x)) = x$  <sup>T5</sup>1c. 當  $x$  值無限趨近某數  $a$  的右側 ( $x > a$ ), 函數值趨近某數  $L$ , 則 right-hand limit =  $L$  <sup>T5</sup>1d.  $f(a)$  defined,  $\lim_{x \rightarrow a} f(x)$  existed,  $\lim_{x \rightarrow a} f(x) = f(a)$  <sup>T5</sup>1e.  $a, b, c \in \mathbb{R}$ , 若  $a < c < b$ .then  $\exists c$  s.t.  $f(a) < f(c) < f(b)$  <sup>T3</sup>1f.  $f, g, h$  is function,  $f < g < h$  when  $x$  around  $a$ ,  $\lim_{x \rightarrow a} f(x) \leq \lim_{x \rightarrow a} g(x) \leq \lim_{x \rightarrow a} h(x)$  <sup>T7</sup>if  $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L$ , then  $\lim_{x \rightarrow a} g(x) = L$ 

2a.  $f(x) = \sqrt{9-x^2}$  <sup>T3</sup>

$9-x^2 \geq 0$

$D = [-3, 3]$

$x^2 \leq 9, -3 \leq x \leq 3$

$R = [0, \infty)$

2b.  $f \circ g(x) = f(x-1) = \frac{1}{\sqrt{x-1}}$  <sup>T6</sup>

$x-1 > 0$

$D = (1, \infty)$

$x > 1$

$R = (0, \infty)$

2c.  $\cos(\tan^{-1}(x))$  <sup>T6</sup>

let  $\theta$ ,  $\tan \theta = x$ , Find  $\cos \theta$ 

$\cos \theta = \frac{1}{\sqrt{x^2+1}}$

$$2e. f(x) = \frac{1}{\sqrt{x}} = (x)^{-\frac{1}{2}}$$

$$f'(x) = -\frac{1}{2} \cdot x^{-\frac{3}{2}} \cdot 1$$

$$= -\frac{1}{2} (x)^{-\frac{3}{2}}$$

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

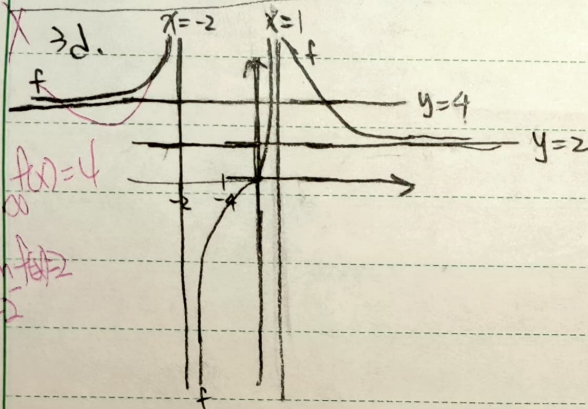
$$f'(a) = -\frac{1}{2} \cdot (a)^{-\frac{3}{2}}$$

$$\begin{aligned} & \text{2b. } \lim_{x \rightarrow -\infty} \frac{3x^3 + 25x^2 + 121}{\sqrt{3x^3 - 5x - 11}} \\ &= \lim_{x \rightarrow -\infty} \frac{3 + \frac{25}{x} + \frac{121}{x^3}}{\sqrt{3 - \frac{5}{x} - \frac{11}{x^3}}} \end{aligned}$$

$$= \frac{\lim_{x \rightarrow -\infty} 3}{\lim_{x \rightarrow -\infty} \sqrt{3}}$$

By LL-01, LL-02,

$$= \frac{3}{\sqrt{3}} = \sqrt{3} \quad \text{By LL-05}$$



$$\lim_{x \rightarrow -\infty} f(x) = 4$$

$$\lim_{x \rightarrow 2} f(x) = 2$$

$$2d. \left[ \lim_{x \rightarrow \frac{\pi}{2}^-} \left( \frac{1}{x} \right) \right] \left[ \lim_{x \rightarrow \frac{\pi}{2}^+} \sin x \right] = \frac{2}{\pi} \cdot \frac{1}{1} = \frac{2}{\pi}$$

$$\rightarrow f: y = \ln x \Leftrightarrow e^y = x$$

$$y = f(x) = \ln(e^x - 3)$$

$$e^y = e^x - 3$$

$$\ln(e^y + 3) = x \ln e = x$$

$$x = \ln(e^y + 3)$$

$$f^{-1}(x) = \ln(e^x + 3)$$

$$e^x - 3 > 0 \quad e^x > 3 \quad x > \ln 3$$

$$f: D = (\ln 3, \infty)$$

$$R = \mathbb{R}$$

$$f^{-1}: D = \mathbb{R}$$

$$R = (\ln 3, \infty)$$

$$e^y + 3 > 0$$

$$e^y = e^x + 3$$

$$2b. c = \frac{p}{q}, p, q \in \mathbb{Z}$$

$$\left| \frac{p}{q} - d \right| \leq \frac{1}{10^n}$$

$$-\frac{1}{10^n} \leq \left( \frac{p}{q} - d \right) \leq \frac{1}{10^n}$$

$$-\frac{1}{10^n} - \frac{p}{q} \leq (-d) \leq \frac{1}{10^n} - \frac{p}{q}$$

$$\frac{1}{10^n} + \frac{p}{q} \geq d \geq -\frac{1}{10^n} + \frac{p}{q}$$

$\Rightarrow \frac{p}{q} \pm \frac{1}{10^n}$  是有理数 By 有理数封闭性

$\exists d$  为有理数 By 有理数稠密性

$$4a. a = \frac{p}{q}, p, q \in \mathbb{Z}$$

$$\left| \frac{p}{q} - b \right| \leq \frac{1}{10^n}, \quad -\frac{1}{10^n} \leq \frac{p}{q} - b \leq \frac{1}{10^n}$$

$$\frac{1}{10^n} + \frac{p}{q} \geq b \geq -\frac{1}{10^n} + \frac{p}{q}$$

$\Rightarrow \frac{p}{q} \pm \frac{1}{10^n}$  是有理数, By 封闭性

$\exists d$  为无理数

$$3a. \frac{\sqrt{h^2+12}-2\sqrt{3}}{h^2} = \frac{\cancel{h^2+12}-12}{h^2(\sqrt{h^2+12}+2\sqrt{3})} = \frac{1}{\sqrt{h^2+12}+2\sqrt{3}}$$

+8

$$\lim_{h \rightarrow 0} \frac{1}{\sqrt{h^2+12}+2\sqrt{3}} = \frac{\lim_{h \rightarrow 0} 1}{\lim_{h \rightarrow 0} (\sqrt{h^2+12}) + \lim_{h \rightarrow 0} 2\sqrt{3}}$$

By LL-01, LL-01, LL-05

$$= \frac{1}{2\sqrt{3}+2\sqrt{3}} = \boxed{\frac{1}{4\sqrt{3}}}$$

$$4c. \forall \epsilon > 0, \exists \delta > 0 \text{ s.t. } 0 < |x-a| < \delta \Rightarrow |f(x)-L| < \epsilon$$

+10

$0 < \epsilon < 1$  From 4a, 4b:  $\lim_{x \rightarrow 0} f(x)$  可能为有理数或无理数, 不唯一

$\Rightarrow \lim_{x \rightarrow 0} f(x)$  DNE,

4d:  $\lim_{x \rightarrow 0} f(x)$  DNE,

没有证明

$\Rightarrow f(x)$  discontinuous at 0

$\Rightarrow f(x)$  is NOT a continuous function

$$\forall \epsilon: 0 < \epsilon < \sqrt{2}$$

$$f(0) < f(a) < f(\sqrt{2})$$

$$0 < f(a) < 1$$

$$f(a) \text{ DNE.}$$

→ failed on IVT.

⇒  $f(x)$  is NOT a continuous function.

$$3C. \lim_{x \rightarrow \infty} \frac{\sin x}{x}$$

| $x$             | $\sin x$             | $\frac{\sin x}{x}$      |
|-----------------|----------------------|-------------------------|
| $\pi$           | 0                    | 0                       |
| $2\pi$          | 0                    | 0                       |
| $\frac{\pi}{2}$ | 1                    | $\frac{2}{\pi}$         |
| $\frac{\pi}{4}$ | $\frac{\sqrt{2}}{2}$ | $\frac{4\sqrt{2}}{\pi}$ |

$\frac{\sin x}{x}$  振荡 不收敛

$$\Rightarrow \lim_{x \rightarrow \infty} \frac{\sin x}{x} \text{ DNE.}$$

$$-1 \leq \sin x \leq 1$$

$$-\frac{1}{x} \leq \frac{\sin x}{x} \leq \frac{1}{x}$$

$$\exists \epsilon. \lim_{x \rightarrow -\infty} \frac{1}{x} = L$$

$$\forall \epsilon > 0, \exists N \text{ s.t. } x < N \Rightarrow \left| \frac{1}{x} - L \right| < \epsilon$$

$$\frac{1}{x} > \frac{1}{N}$$

$$-\epsilon < \frac{1}{x} - L < \epsilon$$

$$-\epsilon + L < \frac{1}{x} < \epsilon + L$$

$$\frac{1}{x} > \frac{1}{N}$$

tautology